47202 Introduction to Future Energy - Ordinary exam - Fall 2024

Der anvendes en scoringsalgoritme, som er baseret på "One best answer"

Dette betyder følgende:
Der er altid netop ét svar som er mere rigtigt end de andre
Studerende kan kun vælge ét svar per spørgsmål
Hvert rigtigt svar giver 1 point
Hvert forkert svar giver 0 point (der benyttes IKKE negative point)

The following approach to scoring responses is implemented and is based on "One best answer"
There is always only one correct answer – a response that is more correct than the rest
Students are only able to select one answer per question
Every correct answer corresponds to 1 point
Every incorrect answer corresponds to 0 points (incorrect answers do not result in subtraction of points)

Q1. Introduction i.

The earth absorbes photons from the sun and emits photons back into space. Which of the following statements best describes the current situation?

Vælg en svarmulighed

- A: The earth emits almost exactly the same number of photons as it absorbes.
- B: The earth emits substantially fewer photons than it absorbes.
- C: The earth and the sun are in balance and emit the same.
- D: The earth emits roughly twenty times fewer photons than it absorbes.
- E: The earth emits roughly twenty times more photons than it absorbes.

Q2. Introduction ii.

As a global average, what is the primary power consumption per person (all energy - not just electricity) roughly?

Vælg en svarmulighed

- A: 230 W
- B: 2.3 kW
- C: 23 kW
- D: 230kW
- E: 2.3 MW

Q3. Wind i.

The largest wind turbine installed at the Østerild test center in Denmark has been announced to have a rated power of P = 21 MW and a rotor diameter of D_{rotor} = 276 m. The wind turbine will be able to supply 25000 Danish households with electricity for a year. If the average electricity consumption is 4 MWh per household per year what is the expected capacity factor(CF) of the wind turbine?

Vælg en svarmulighed

- A: 20-30 %
- B: 30-40 %
- C: 40-50 %
- D: 50-60 %
- E: 60-70 %

Q4. Wind ii.

If the large wind turbine installed at the Østerild test center with a rated power of P = 21 MW and a rotor diameter of D_{rotor} = 276 m is reduced to have a rotor of half the diameter what is the expected rated power of the turbine?

Vælg en svarmulighed

- A: 0-3 MW
- B: 3-6 MW
- C: 6-9 MW
- D: 9-12 MW
- E: 12-15 MW

Q5. Biomass i.
Concerning green methanol. What statement is not true?

Vælg en svarmulighed

- A: Biomass can be the carbon source for the production
- B: CO2 can be the carbon source for the production
- C: It can be produced from syngas
- D: Anaerobic digestion can be used to produce syngas
- E: Methanol can be used as a fuel in the transport sector

Q6. Biomass ii. Consider methanol production from the following syngas: 60 mol% H2 and 25 mol% of CO and 15% CO2. How much methanol can potentially be produced if you have a syngas flow of 0.3 kmol/s? LHV_methanol = 638 MJ/kmol.

Vælg en svarmulighed

- A: 48 MW of methanol
- B: 54 MW of methanol
- C: 58 MW of methanol
- D: 64 MW of methanol
- E: 68 MW of methanol

Q7. Nuclear i.
Plutonium is formed in light water reactors by

Vælg en svarmulighed

- A: Neutron absorption on ^{235}U
- B: Neutron absorption on ^{238}U
- C: Fission of ^{235}U
- D: Fission of ^{238}U
- E: Adding thorium to the fuel

Q8. Nuclear ii.
Water outlet temperature in a pressurized water reactor is approx.

Vælg en svarmulighed

- A: 220 ^{o}C
- B: 275 ^{o}C
- C: 315 ^{o}C
- D: 450 ^{o}C
- E: 520 ^{o}C

Q9. Solar i.
Which of the following statements best describes a solar cell based on a semiconductor with a bandgap of Eg, (assuming the Eg is in the range 0.5 eV to 3.5 eV)?

Vælg en svarmulighed

- A: The open circuit voltage, Uoc under solar illumination is given by: Uoc = a* Eg where a is a constant.
- B: The short circuit current, Isc under solar illumination is given by: Isc = a* Eg , where a is a constant.
- C: The fill factor, FF under solar illumination is given by: FF = a* Eg + b, where a and b are constants.
- D: The short circuit current, Isc under solar illumination is given by: Isc = a* Eg + b, where a and b are constants.
- E: The open circuit voltage, Uoc under solar illumination is given by: Uoc = a* Eg + b, where a and b are constants.

Q10. Solar ii.

What are examples of "soft costs" in the context of solar installation costs?

Vælg en svarmulighed

- A: inverters, electronics
- B: permitting, customer acquisition, legal expenses
- C: solar panels, mounting hardware (such as racks)
- D: power transformers, high voltage cabling
- E: costs to make foundations, if the ground is too soft for installing racks directly

Q11. Water i.
Which one of the following technologies utilizes the chemical energy stored in water?

Vælg en svarmulighed

- A: Hydro power
- B: Wave power
- C: Osmotic power
- D: Geothermal energy
- E: Tidal power/ocean current power

Q12. Water ii.

The Itaipu hydropower station at the border between Brazil and Paraguay has a nominal power of 14 GW. In 2016 it produced 103.1 TWh electricity. What is then capacity factor of the Itaipu hydropower station for the year of 2016?

Vælg en svarmulighed

- A: ~20%
- B: ~40%
- C: ~60%
- D: ~80%
- E: ~100%

Q13. Thermodynamics and electrochemistry i.
Gasoline is a mixture of many species, so very often octane (C_8H_{18}) is used as a model substance for gasoline. Octane is combusted by the following reaction:

$C_8H_{18} + 25O_2 \rightarrow 8CO_2 + 9H_2O$

Calculate the enthalpy of combustion of octane from the thermodynamic values in the table below:

	C_8H_{16}	C	H_2	O_2	CO_2	H_2O
Δh_f (kJ/mol)	-250.1	0	0	0	-393.5	-285.8
Δg_f (kJ/mol)	6.4	0	0	0	-394.4	-237.1

Vælg en svarmulighed

- A: -5470.1 kJ/mol
- B: -5295.5 kJ/mol
- C: -429.2 kJ/mol
- D: 429.5 kJ/mol
- E: 5470.1 kJ/mol

Q14. Thermodynamics and electrochemistry ii.
The silver oxide battery is a well-known commercial battery used as button cells in watches. The cell voltage is of course known, but if it was not, it can be calculated form the thermodynamic functions of the involved species.

The overall reaction is $Zn + Ag_2O \longrightarrow ZnO + 2Ag$ (two electrons are exchanged)

Enthalpies of formation and Gibbs free energies of formation

	Zn	Ag_2O	ZnO	Ag
Δh_f (kJ/mol)	0	-31.1	-350.5	0
Δg_f (kJ/mol)	0	-11.2	-320.5	0

Calculate the theoretical cell voltage of the silver-oxide battery.

Vælg en svarmulighed

- A: 1.45 V
- B: 1.50 V
- C: 1.55 V
- D: 1.60 V
- E: 1.66 V

Q15. Fuel cells and hydrogen i.
A fuel cell stack is made from 250 cells, each with the cell area 200 cm^2. The cells are operated at 1.5 A/cm^2 and a cell voltage of 0.7 V. Calculate the total voltage, total current and total electrical power of the stack.

Vælg en svarmulighed

- A: 140 V; 1.5 A; 210 W
- B: 140 V; 300 A; 42 kW
- C: 175 V; 300 A; 52.5 kW
- D: 175 V; 375 A; 65.6 kW
- E: 35,000 V; 1.5 A; 52.5 kW

Q16. Fuel cells and hydrogen ii.
Is a fuel cell limited by Carnot's law?

Vælg en svarmulighed

- A: Yes, Carnot's law describes a general limitation for extracting work from a flow of heat or fuel.
- B: Yes, Carnot's law is the ultimate thermodynamic limit of any work producing process.
- C: It might be limited by carnot's law. It depends on the fuel and if there is a net destruction of entropy due to a net reduction in the number of gas molecules.
- D: No, A fuel cell does not produce work, only electrical energy directly and not via a state of work.
- E: No, a fuel cell does not operate via a heat to work process and therefore it is not limited by Carnot's law

Q17. Power-to-X i.
Which of the following capabilities are part of the purpose of the electrolyte in an electrolyzer cell:
(1) Preventing the produced gases from mixing
(2) Preventing the electrodes from touching each other
(3) Conducting ions between the electrodes
(4) Conducting electrons between the electrodes
(5) Conducting gases between the electrodes

Vælg en svarmulighed

- A: (1); (2); (3)
- B: (1); (3); (4)
- C: (1); (2); (4)
- D: (2); (3); (4)
- E: (2); (3); (5)

Q18. Power-to-X ii.

An electrolyzer stack with 300 cells is fed with liquid water and produces hydrogen at a total voltage of 550 V and a total current of 800 A. Which one of the following statements is most correct:

Vælg en svarmulighed

- A: The stack consumes 440 kW electricity, and a significant part of that is emitted as heat.
- B: The stack is consuming a significant amount of heat, because water electrolysis is always a net heat consuming process.
- C: The stack consumes about 440 kW electricity. It may be a net heat consumer or a net heat producer. We cannot tell from the information available.
- D: The stack is thermally balanced and thus it is neither a net heat producer or a net heat consumer.
- E: First law says that energy cannot disappear. The 440 kW electricity that the stack consumes is fully converted into heat along with the hydrogen that is produced.

Q19. Storage i.
Considering the present status of technological progress, which one of the following has the highest mass energy density (per unit of mass, i.e. per kg)?

Vælg en svarmulighed

- A: Lithium ion battery
- B: Gasoline
- C: Natural gas
- D: Liquid hydrogen
- E: Ethanol

Q20. Storage ii.
Considering the present status of technological progress, which one of the following has the highest peak power density (in W/kg)?

Vælg en svarmulighed

- A: Super capacitor
- B: Methanol
- C: Hydrogen fuel cell
- D: Gasoline
- E: Zn-air battery

Q21. Batteries i.

An old battery chemistry has recently made a reemergence on the battery market for automotive applications. Which chemistry is it, and what it the main reason for its success?

Vælg en svarmulighed

- A: Lithium-ion phosphate (LFP) batteries because of their low cost
- B: Na-ion batteries because of their higher energy density than Li-ion
- C: Ni-Cd batteries because of their low toxicity
- D: All-solid-state batteries because of their long lifetime
- E: Vanadium redox flow batteries because of their high voltage

Q22. Batteries ii.

A lemon battery consists of two electrodes based on zinc with a half-cell potential of -0.76 V, and copper with a half-cell potential of 0.34 V, and Faraday's constant is F= 96485 C/mol.

Which statement is most correct?

Vælg en svarmulighed

- A: Zinc is the negative electrode, and it is reduced during discharge
- B: The open circuit voltage (OCV) is -0.42 V
- C: Copper is the negative electrode and is reduced during discharge
- D: The Gibbs free energy of reaction is ΔG = -212.3 kJ/mol
- E: Neither zinc nor copper is oxidized during discharge, but the lemon juice is

Q23. Infrastructure i.
(Fermi) Estimate the number of piano tuners in Copenhagen. Calculate log10(number of piano tuners in Copenhagen) (i.e. take the base 10 logarithm of your estimate) and pick the closest number:

Vælg en svarmulighed

- A: 0 (1 tuner)
- B: 1 (10 tuners)
- C: 2 (100 tuners)
- D: 3 (1000 tuners)
- E: 4 (10000 tuners)

Q24. Infrastructure ii.
Which of the following statements best describes the LCOE (levelized cost of energy)?

Vælg en svarmulighed

- A: Discounted costs divided by undiscounted energy flows. Takes externalities into account.
- B: Discounted energy flows divided undiscounted costs . Does not take externalities into account.
- C: Discounted energy flows divided discounted costs . Takes externalities into account.
- D: Discounted costs divided by discounted energy flows. Does not take externalities into account.
- E: Discounted energy flows divided discounted costs . Does not take externalities into account.